Sprint 4: Network Security Hardening and Improvements

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Overview

Performed additional network security hardening and configuration improvements. The objective was to strengthen the network's security posture, improve consistency with the established configuration baseline, and address remaining hardening recommendations ahead of the MVP demonstration.

Improvements were made across all eight network devices, including R1, R2, MLS1, MLS2, and SW1–SW4.

1. Router Security Improvements

Reviewed R1 and R2 and brought their administrative and security configurations up to date with the established network baseline.

- Updated missing SSH, AAA, RADIUS, local authentication, and management-access security settings.
- Standardized administrative session timeouts, login protections, and SSH authentication retries.
- Implemented external anti-spoofing protection on R1.
- Disabled unnecessary CDP advertisements on R1's external-facing interface.
- Applied supported interface-hardening settings, including disabling Proxy ARP.
- Preserved existing NAT/PAT functionality, addressing, and routing configurations.

2. OSPF Security Improvements

Implemented passive-interface hardening across R1, R2, MLS1, and MLS2.

- Configured passive-interface default to prevent unnecessary OSPF neighbor discovery.
- Explicitly permitted OSPF activity on required routing interfaces.
- Preserved the routing relationships between the routers and multilayer switches.
- Verified that required OSPF adjacencies returned to FULL status.
- Confirmed departmental VLAN networks remained advertised and reachable.

These improvements reduced unnecessary routing protocol exposure without changing the established network topology.

3. Access-Layer Security Improvements

Implemented additional protections across SW1–SW4.

- Disabled CDP on endpoint-facing interfaces to reduce unnecessary device information exposure.
- Configured broadcast storm control at a 5% threshold on applicable PC and server ports.
- Reviewed existing port security, DHCP Snooping, PortFast, BPDU Guard, and trunk configurations.
- Maintained appropriate protections on endpoint ports while preserving separate configuration requirements for wireless infrastructure connections.

4. Management Security Standardization

Improved consistency across all eight network devices.

- Standardized localadmin accounts with privilege level 15.
- Configured SSH authentication retries to three attempts.
- Enabled login blocking on R1, R2, MLS1, and MLS2 where supported.
- Disabled unnecessary CLI DNS lookups using no ip domain-lookup.
- Confirmed SSH-only remote administration and existing management ACL protections.

The access switches did not support the login-blocking command, which was recorded as a platform limitation.

5. Configuration Limitations and Follow-Up

Several optional hardening features were unsupported by Packet Tracer, including certain ICMP redirect, ACL logging, and login-blocking commands.

A BPDU Guard configuration attempted on SW4's WLC-facing port also triggered an error-disabled condition. This identified an important device-specific exception to the standard endpoint-port hardening baseline. The WLC and AP connections should retain configurations appropriate to their infrastructure roles.

Remaining minor configuration cleanups were deferred to avoid unnecessary disruption to the established network.

Outcome

Completed the major planned security improvements across the network and reviewed configurations for consistency with the project baseline.

The improvements strengthened administrative access, routing protocol security, access-layer protection, and external-interface filtering while preserving the existing network architecture.

Status: Sprint 4 security hardening improvements completed; final testing and QA remain in progress.